

AI Search Algorithms in Games: Homework

2

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Homework - Game 1: Branching Factor

(A) **How many different/distinct moves can the "red" player make?** Based on the board situation provided, the red player can make a total of 44 distinct moves. The breakdown is as follows:

- **5 Soldiers** : 5 moves (each can only move forward 1 step).
- **2 Chariots** : 2 moves (each is mostly blocked but has 1 forward valid move).
- **2 Knights** : 4 moves (each has 2 forward jumps available).
- **2 Elephants** : 4 moves (each has 2 valid diagonal jumps).
- **2 Advisors** : 2 moves (each can move to the center).
- **1 General** : 1 move (can step forward).
- **2 Cannons** : $13 + 13 = 26$ moves (horizontal sliding and vertical jumps/slides combined).

Total Branching Factor (b) = $5 + 2 + 4 + 4 + 2 + 1 + 26 = 44$.

(B) **1-step look-ahead strategy evaluation count:** Using a 1-step look-ahead, the computer only evaluates the immediate children of the current state. With a branching factor of 44, the computer needs to do **44** heuristic evaluations.

(C) **3-step look-ahead strategy evaluation count:** Using a 3-step look-ahead strategy, the evaluation requires looking at the sequence of (Player Move $\rightarrow$ Opponent Move $\rightarrow$ Player Move). Assuming an average branching factor of 44 for both sides, the number of leaf node evaluations is:

$$44 \times 44 \times 44 = 44^3 = 85,184 \text{ evaluations.}$$

Homework - Game 2: Tic-Tac-Toe Completions

Here are the simulated completions for the top two incomplete branches using natural heuristics:

1. **Top Branch:** Turn 1: X center → Turn 2: O right edge → Turn 3: X top-right → Turn 4: O bottom-right → Turn 5: X top-left → Turn 6: O bottom-left → Turn 7: X top-mid. **Result: X wins.**
2. **Second Branch:** Turn 1: X center → Turn 2: O bottom-right → Turn 3: X top-left → Turn 4: O bottom-mid → Turn 5: X top-mid → Turn 6: O top-right → Turn 7: X mid-left → Turn 8: O mid-right → Turn 9: X bottom-left. **Result: Tie.**

Homework - Game 3: Value Backup for 1st Move

By performing value backup from the bottom layer (terminal states) of the Tic-Tac-Toe search tree up to the root node using the Minimax algorithm, the backed-up utility value for an empty board is **0**. This indicates that with optimal play from both MAX (X) and MIN (O), the game will inevitably end in a draw, regardless of whether the 1st X is placed in the center, corner, or edge.

Homework - Game 4: 2-Step-Ahead Search

Based on the provided 2-step-ahead search tree and the heuristic evaluation function $E(n) = M(n) - O(n)$: The backed-up MIN scores for MAX's possible moves are evaluated. Placing X in the **Center** yields the highest guaranteed heuristic score of **2** after minimizing the opponent's responses. Placing X in other locations (like the top-right) yields lower guaranteed scores (e.g., 1). Therefore, **putting X in the center** is the best move.

Homework - Game 5: Tree Properties

1. **Leaf nodes occur at levels 5, 6, 7, 8, 9. Why?**
A minimum of 5 moves is required to win Tic-Tac-Toe (3 moves for X, 2 for O). Therefore, no game can terminate before level 5. A full board without a winner terminates at level 9. Thus, all leaf nodes must exist between these levels.
2. **Leaf nodes on level 5 have value +1. Why?**
Level 5 is reached immediately after X's 3rd move. Because X just moved, only X can complete a winning line at this depth. Since X is MAX, the utility is +1.
3. **Leaf nodes on level 6 have value -1. Why?**
Level 6 is reached immediately after O's 3rd move. Only O can complete a winning line at this turn. Since O is MIN, the utility is -1.
4. **Leaf nodes on level 9 have values 0 or +1. Why?**
Level 9 is the final move of the game, played by X. If X completes a line, the value is +1. If X does not complete a line, the board is full and the game is a draw, yielding a value of 0. It cannot be -1 because it is X's turn, not O's.
5. **How many leaf nodes on level 5?**
There are **1,440** leaf nodes at level 5. (Calculated as 8 winning lines × 3! ways to place X × 6 × 5/2 × 2! ways to place O).

6. **How many leaf nodes on level 6?**

There are **5,328** leaf nodes at level 6.

7. **How many leaf nodes on level 9?**

There are **127,872** leaf nodes at level 9 (combining all end-game wins for X on the final move and all full-board draws).